

Standard Form — Writing and Converting

AQA · KS4 Year 9–10

Common Mistakes to Avoid:

- A must satisfy $1 < A < 10$. So 36×10^4 is wrong; correct: 3.6×10^5 .
- Positive power = big; negative power = small.
- Count places carefully — 0.00047 needs 4 moves 4.7×10^{-4} .
- $0.003 = 3 \times 10^{-3}$, NOT 3×10^3 .

Method Steps:

1. 1 Move decimal so $1 < A < 10$
2. 2 Count places moved = power n
3. 3 Big number (> 1) positive n
4. 4 Small number (< 1) negative n
5. 5 Check by converting back

Worked Example:

Write 0.000 47 in standard form.

Step 1 — Move decimal 4 places: 4.7

Step 2 — Original < 1 , so $n = -4$

Answer: 4.7×10^{-4}

Visual Aid: Place-value slide showing the decimal point moving left for big numbers (positive power) and right for small numbers (negative power). Arrow diagram: 5,600,000 $\rightarrow$ 5.6 (6 places left) = 5.6×10^6 . And 0.0034 $\rightarrow$ 3.4 (3 places right) = 3.4×10^{-3} .

Question 8 [5 marks]

Bacterium = 2×10^{-6} m; virus = 3×10^{-8} m. a) How many times longer? b) 500 bacteria end-to-end? c) Viruses in 1 mm?
